

Vince:

You may be interested in the following interview which appeared on ERisk.com last Friday. Were Rick Buy's comments about real options taken out of context?

Yann Bonduelle leads a 25-person team for PricewaterhouseCoopers in London that applies decision analytics and real options theory to dilemmas ranging from valuing a biotechnology product to deciding whether to kill off an Internet financial services business. Here he talks to Rob Jameson about whether this "theoretical" approach to risky decision-making really helps businesses in their day-to-day balancing of risk and reward. Yann holds a Ph.D degree from the Engineering-Economic Systems Department at Stanford University, where he studied how to apply engineering decision and design analysis to wider economic, social and business issues. He then worked as a consultant applying his decision analysis methodologies to problems that included consumer decision making about innovative products such as electrical vehicles, before joining the PricewaterhouseCoopers team in 1998 where he is now a partner. He has written widely on the application of real options, particularly in fields of life sciences, technology, and e-business, and has a special interest in the relationship between risk assessment, validation of risk data and financial valuation.

How would you sum up your approach to business decision analytics?

Most of our projects are set up to help businesses that face massive uncertainties of some kind. Decision analysis helps people explore problems, and redesign their decision-making process to increase the chance of them making the right choices. For example, imagine a biotechnology company that has to decide whether to put itself up for sale, enter a strategic relationship, or continue to go it alone. Each of those options will lead on to other value-enhancing or value-destroying scenarios. We work with the client firstly to understand and challenge the assumptions associated with their most likely business development scenarios, and secondly to help them identify decisions that would help protect or increase the value of their technology or company. Quantifying technical, regulatory or commercial risks can sometimes be a challenge. In technology-intensive fields, however, we have found that managers (often scientists) are quite willing to describe the main sources of risk and to assess the probability that a risky event may or may not occur.

How does this kind of risky decision-making relate to real options valuation?

You can't say what the value of an asset is until you decide what you might use it for. This means that, to form an opinion about the value of an asset, you must explore the most important decisions that you are likely to face and that will have a significant impact on the value of the asset. So decision analysis helps to define the business problem and to uncover a stream of inter-related choices that are, in effect, "real options". For example, if a company is trying to decide whether to invest in a risky project, does it have the option to pull the plug on the investment at an early stage if a pilot project gives a poor showing? That "real" option reduces the riskiness, and increases the potential value, of the original business plan. So real options and decision analysis are really very close to one another. But you don't have to believe in real options valuations to find decision analysis useful.

What do you mean?

Often decision analysis can help managers to identify the key risks in a strategic decision, attach weights to these, and show clearly how they interact. For many companies this "risk discovery" is the most valuable part of the exercise.

Real options theory has been criticised recently for being, well, not very realistic. Is it a practical approach to valuation?

It's important not to hold out unrealistic hopes for the real options approach to valuation. But it's an exciting methodology, and it's also sometimes the only reasonable way of tackling a very practical problem. For example, when a firm sells an asset, the firm might have to make an independent valuation of the asset for legal or corporate governance reasons. But in many businesses today there are assets that simply cannot be valued in traditional ways because they are difficult to link to cashflows. The cashflows might not exist because the business is so novel, or they might be hidden. In some respects, a real options analysis is much closer to reality than a traditional valuation.

How, exactly?

The classic way of valuing a future business is to base the calculation on a single discounted cashflow that is projected from the activity. But this doesn't really take account of the way that scenarios can change, or the fact that managers can react to situations as they unfold. I mentioned earlier the option to kill a project or business at an early point. But the upside is that if a pilot project yields exciting results, it might allow you to invest more quickly and reach a revenue-generating position in a much shorter time than the original business plan allows. So to value a future business we really need to look at the cashflows that might arise in a number of scenarios. This is "realistic" in that, if the project gets the green light, you can bet that its managers will be taking that kind of decision on the ground all of the time.

What's the most challenging part of mapping out a decision analysis tree?

Modelling the links between the variables in the decision tree -- it's something we have particular strengths in. But it's also tricky to know when it's worthwhile to add on more detail, and when it's better to draw back.

In a recent ERisk interview, Rick Buy, chief risk officer of Enron, said that over the two years that Enron had experimented with the real options concept, it had found it of "limited, but not zero, use". Why is there a slight air of cynicism about real options in some businesses today?

It's strange that Enron would profess this attitude. A few years ago, it was widely reported to have used real option valuation to support a very profitable purchase decision. They had apparently bought cheaply some older generators in the US that generated electricity at a very high cost. They knew that they could mothball them for most of the year, and switch them on only when the electricity prices were sufficiently high. Nevertheless, from a customer's point of view, there might have been too much hype about the methodology. One problem in the application of real options technology is that there are, perhaps, too many people trying to tweak reality to conform to their "perfect" model. It's better to aim for something pragmatic that clearly improves decisions over time. In one pharmaceutical company we worked with recently, we worked together to improve their valuation analyses by moving from a single discounted cash-flow methodology to one that took into account a rather small set of business scenarios. It would have shocked

some academics and consultants, but it was an undeniable improvement on the original approach.

Why do you think financial institutions are only just picking up on your field, when it's been applied in the energy industry for 15 years or more?

It might have something to do with the relative stability of the banking world until recently, and the relatively high margins that banking lines have enjoyed. Also, industries such as energy and pharmaceuticals tend to have more people with an engineering and science background. The dynamic modelling of decisions is based on methodologies originally dreamed up to help engineers design electrical and electronic systems. This approach is quite distinct from the Black-Scholes options analyses that the banking world is familiar with: the Black-Scholes approach is difficult to apply in a real options context, because everything depends on the assumptions that you put into the Black-Scholes model. The real options approach, on the other hand, is in a sense a way of modelling those assumptions more explicitly. But banks are now adopting some of the thinking, particularly in terms of using decision analysis to pinpoint risks and identify value-enhancing decisions, and in using real options methodologies to sort the wheat from the chaff in their more speculative investments.

You mean their Internet investments?

We have recently worked with a major Dutch bank that had arrived late in the Internet game, and then made a considerable number of investments. Now that even B2B business models have question marks hanging over them, and many B2C businesses are already under water, they wanted to work out which investments might contain real value. In this situation, it's a case of ranking priorities and helping the bank make sense of what could turn into a decision-making chaos, rather than sophisticated valuation. It's not just a case of whether an internet investment should be killed off, but the problem of whether continued funding for it should take priority over budget demands for major IT upgrades in existing businesses, and so on. These are very practical questions and they have to be answered somehow.

Are there other areas in financial institutions that seem accessible to this approach?

Yes, for example, we think it can help work out the value associated with various approaches to marketing a new bank business line. At the moment, many banks are chasing high-net-worth individuals, but it's not always clear which kind of individual a particular bank should decide to pursue. The bank might have a regional or industry advantage already in one particular area, for example, music business people. But what is the churn rate associated with this kind of customer? What is the profitability associated with the customer segment? Will the time and cost benefits of the advantages the bank has in the sector outweigh any disadvantages? Weighing up this kind of complex problem, where one thing leads to and depends on another, is what decision and real options analysis is good at.

Is there any way of rigorously backtesting or validating real options valuations?

In all honesty, not really. The problem is that by the time the option is exercised, many of the variables surrounding it will have changed, so it's difficult to compare our original analysis with how things turn out. However, the value of the analysis

comes not only from the final number ("the value of this asset is X") but also from providing a thorough process, an outsider's point of view, an understanding of the sources of value and, in short, a bit of clearer thinking.

If real options are so important, why are they so rarely cited in communications to shareholders and equity analysts?

The battle is still to convince companies to use real option valuations as a significant part of their internal analysis. Even in major companies in the oil & gas, and pharmaceutical sectors, where the ideas have taken some root internally, there seems to be a lot of reluctance to use them in external communication. We are working with analysts to understand better what they need to if we are to move things on to the next step.

How does the riskiness of a business, in terms of the major strategic dilemmas it faces, relate to its share value, and its capital structure?

That's a big question. It's related to work my colleagues do on the optimal debt-to-equity capital structure and gearing of a corporation, which in turn arises out of the likely revenue and cost volatilities of the business. The more volatile the business, the less gearing it can sustain, and the higher the cost of capital. Our work touches on this in the sense that exercising (many) specific real options can allow a firm to change its nature, and thus also its risk profile. One classic example is the pharmaceutical industry. A host of different kinds of companies service that industry from "big pharma" companies through to smaller biotechnology startups and run-of-the-mill contract research organisations. A contract research organisation is often operating within a very competitive environment with relatively few risks--it does not invest in drug development itself -- but also very thin margins. But in fact, a few of these companies have used the skills and knowledge they have developed to become much more substantial and profitable healthcare companies of various kinds. It's an example of a company exercising the real options that lie within its skills and assets to transform its own identity.

Rob Jameson, ERisk